Practical Examination

Deploy a Student Event Registration Portal

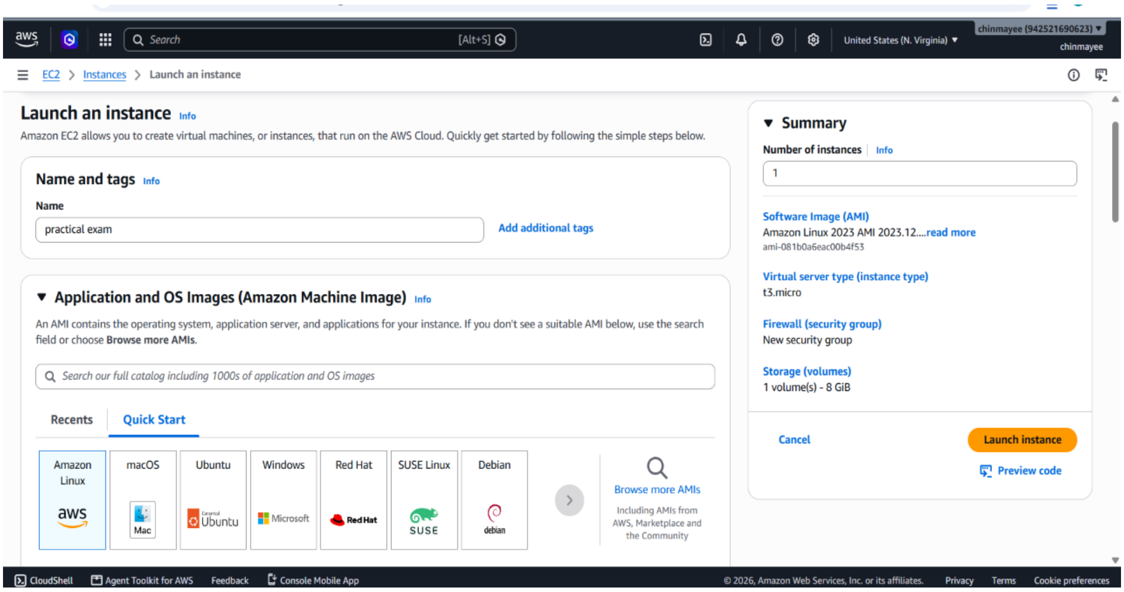
Problem Statement

You have been assigned to deploy a **Student Event Registration Portal** on a Linux server.

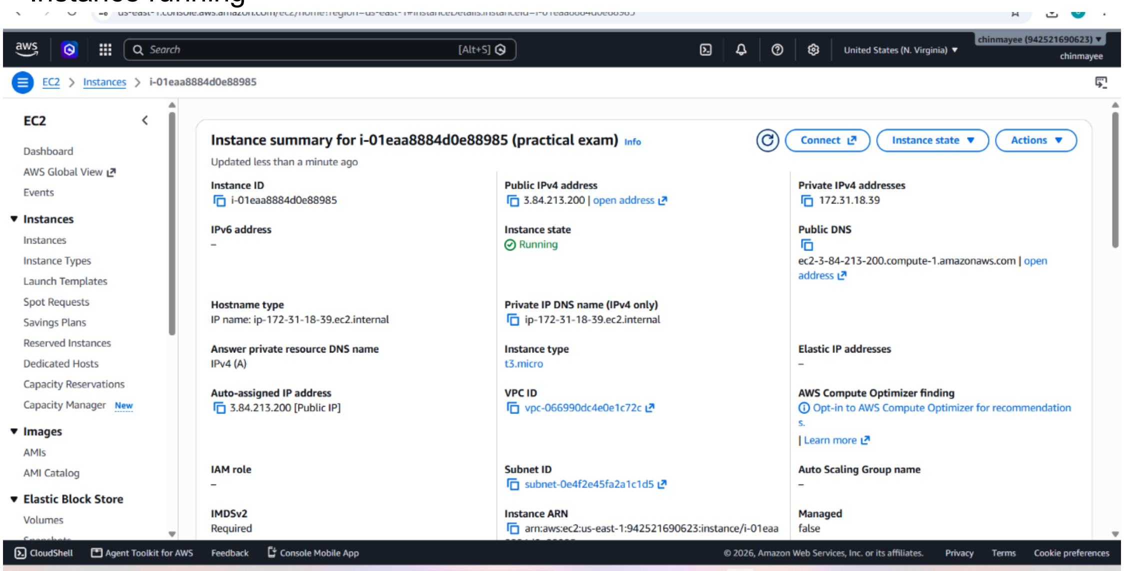
You must build and host a simple dynamic website that allows students to **register for an event and log in using their registered credentials**.

The website must be deployed using **either LAMP or LEMP** and must use **MySQL/MariaDB** for storing user information.

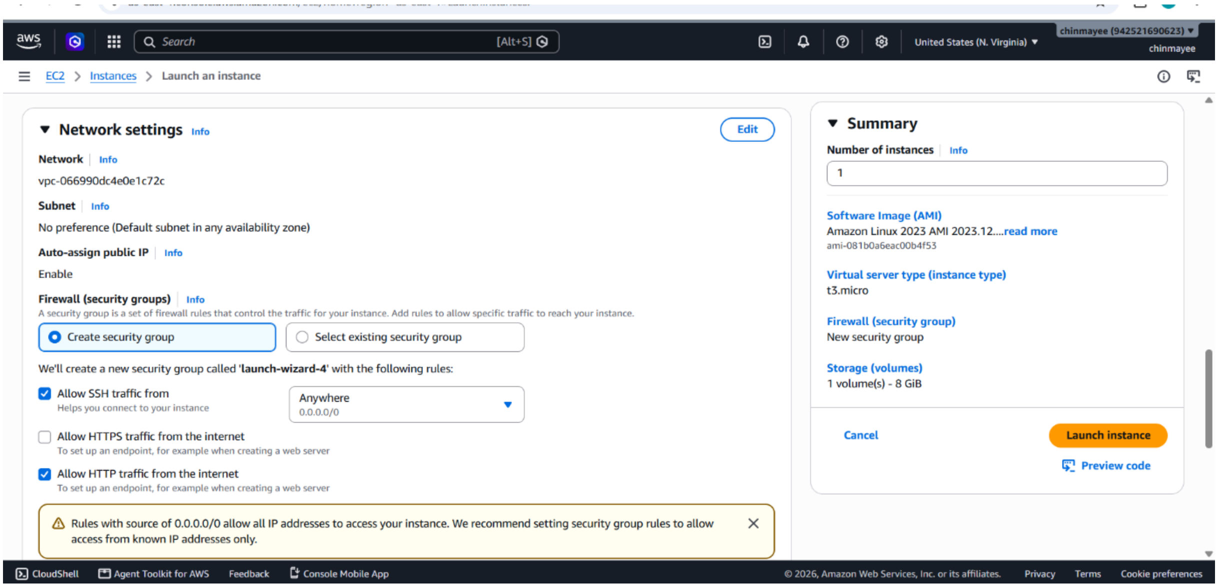
1. **AWS EC2**
   - EC2 instance creation



   - Instance running

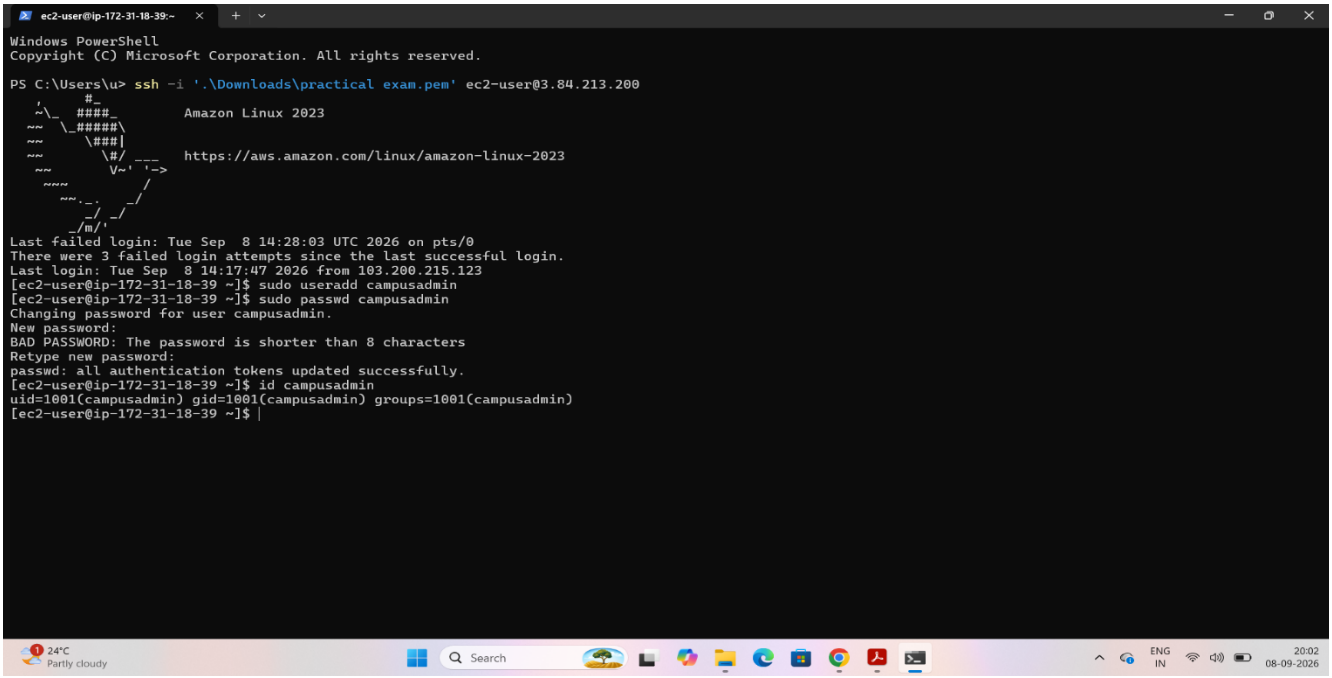


○ Security Group configuration



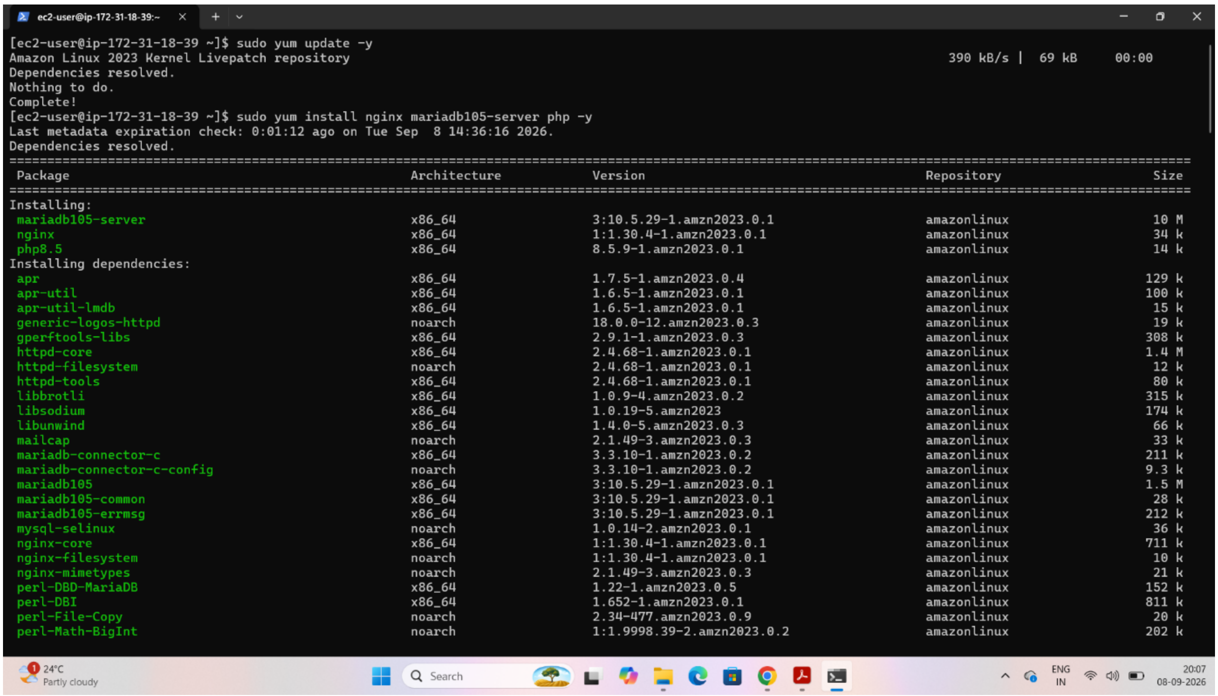
2. **Linux**

○ SSH connection ○ Linux user creation ○ Important Linux commands/results

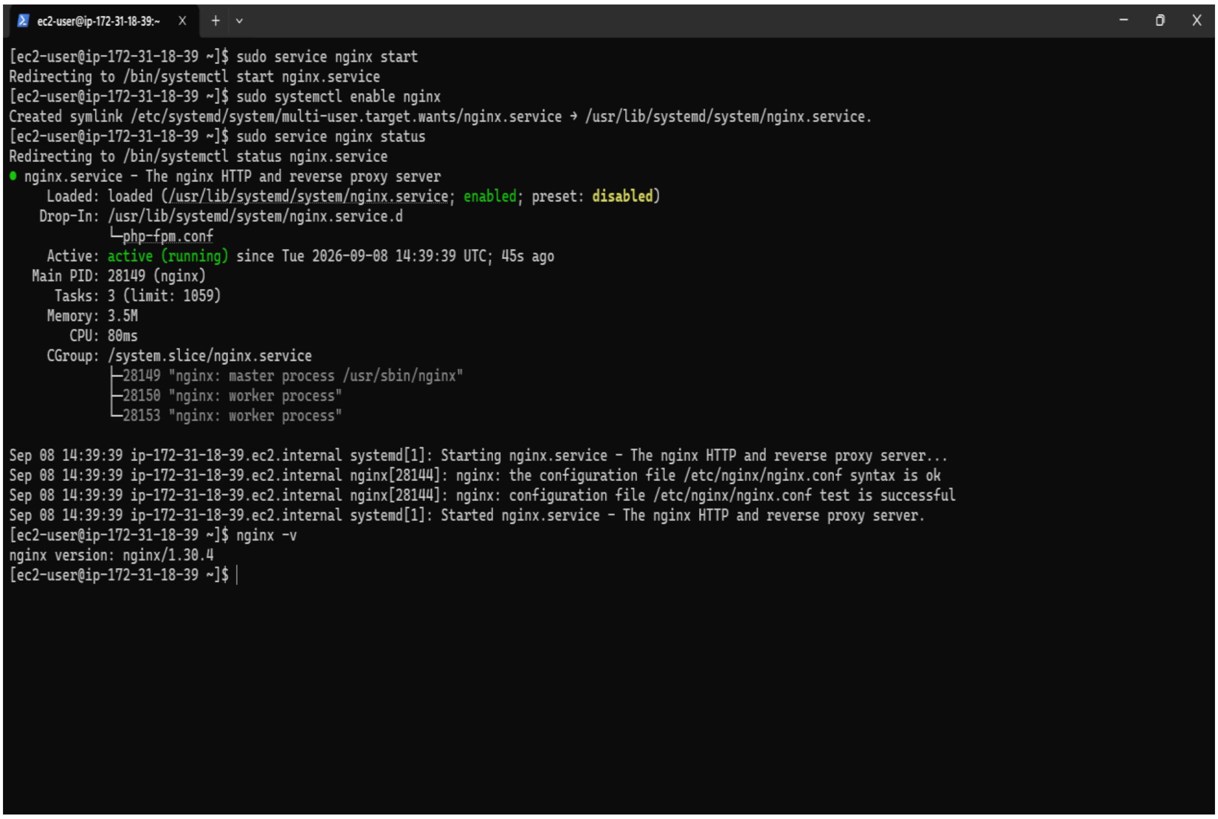


3. **LAMP/LEMP**

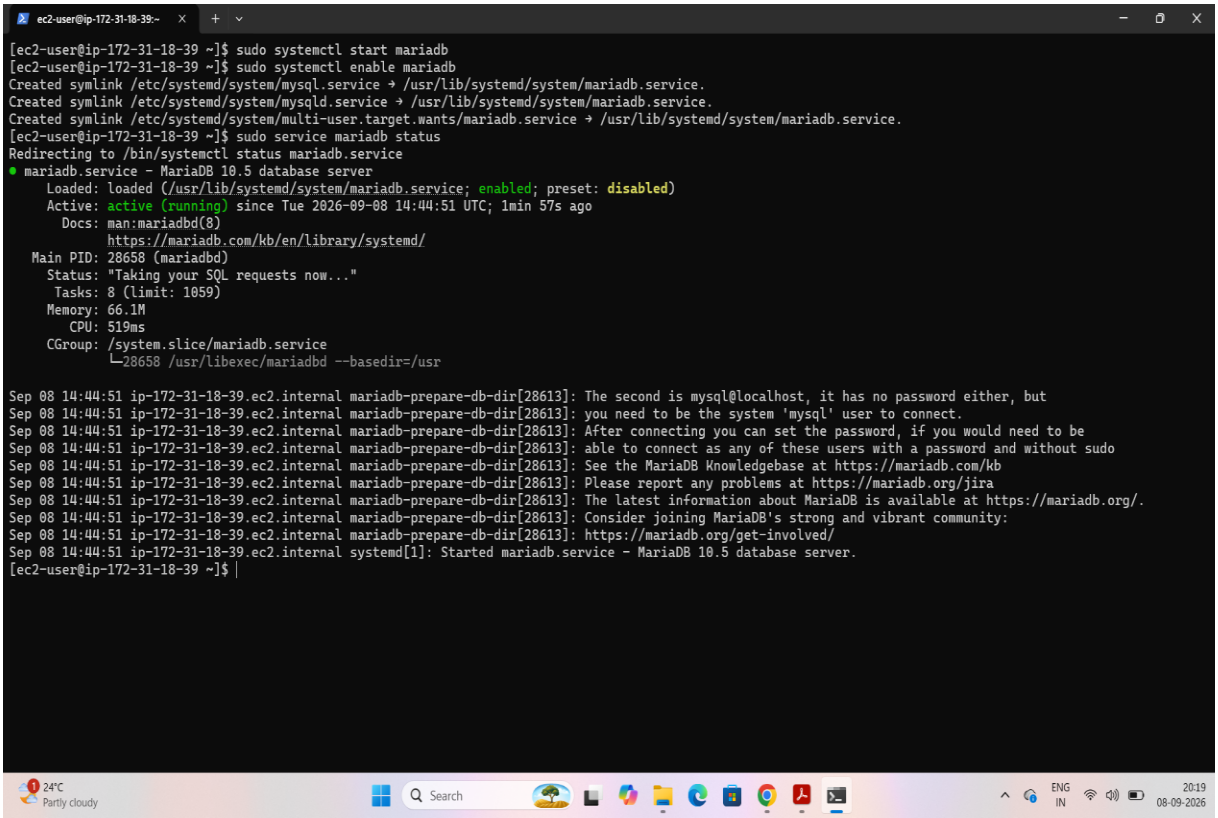
  - Installation



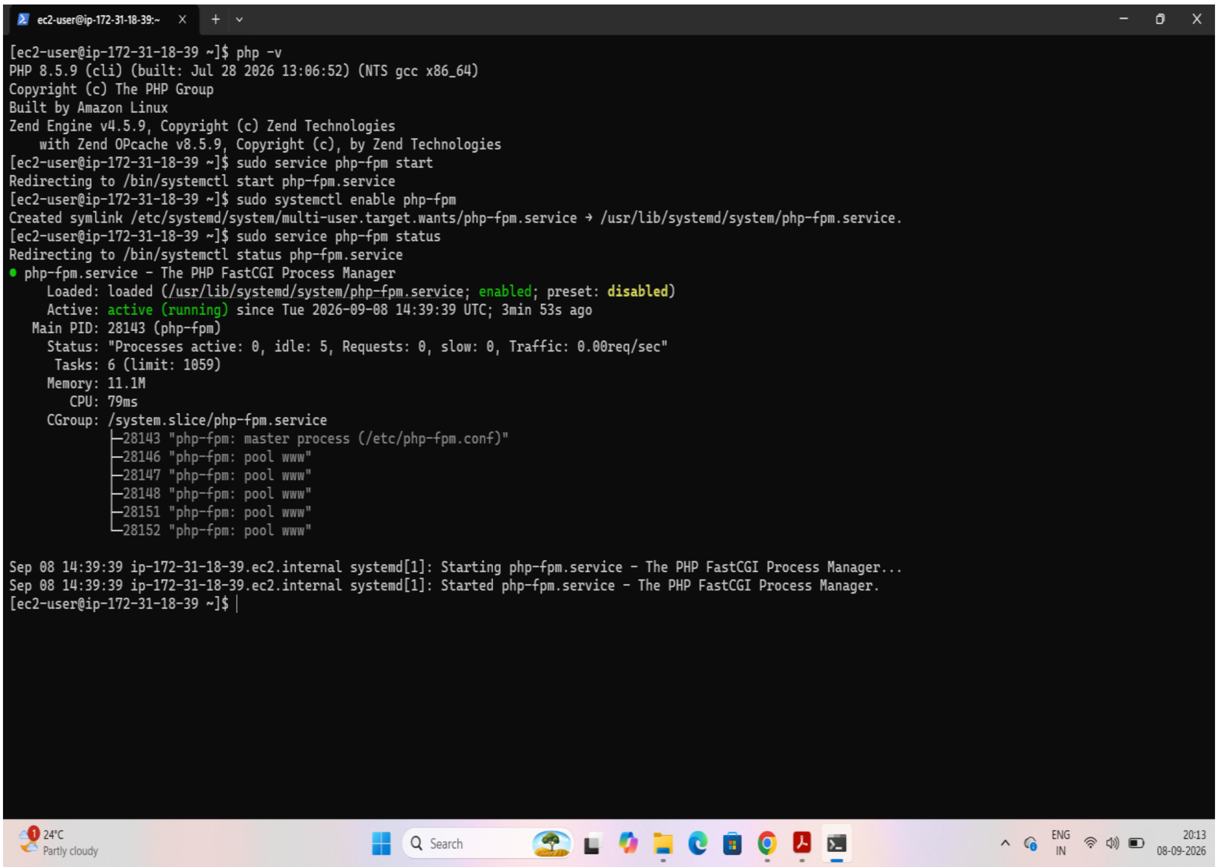
  - Nginx status



○ MySQL/MariaDB status

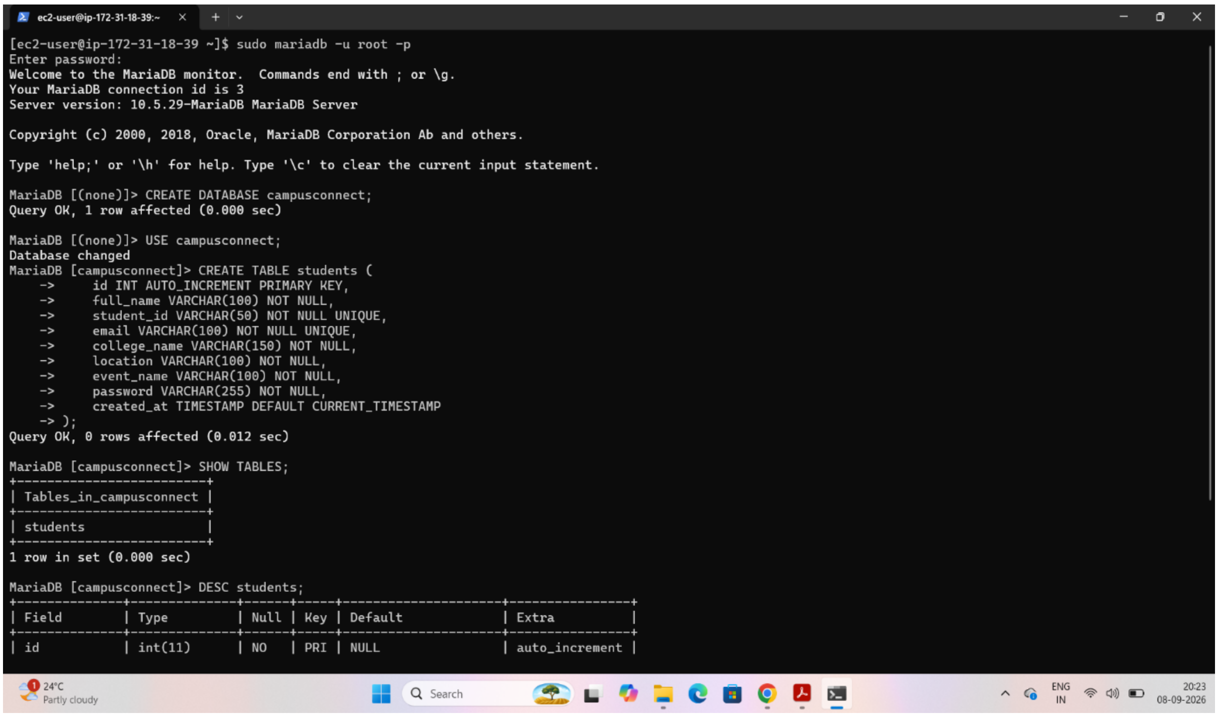


○ PHP installation/version

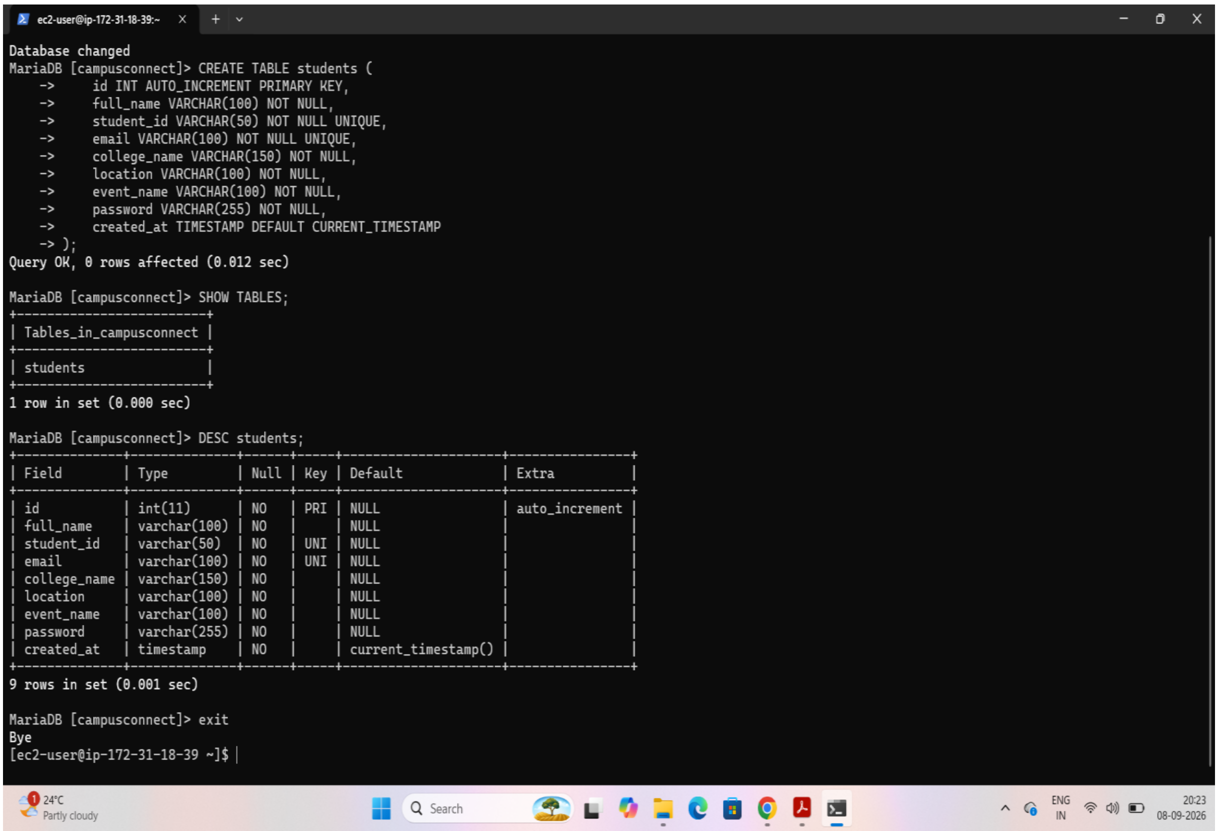


4. **SQL**

   ○ Database creation ○ Table creation



```
[ec2-user@ip-172-31-18-39 ~]$ sudo mariadb -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE campusconnect;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE campusconnect;
Database changed
MariaDB [campusconnect]> CREATE TABLE students (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     full_name VARCHAR(100) NOT NULL,
    ->     student_id VARCHAR(50) NOT NULL UNIQUE,
    ->     email VARCHAR(100) NOT NULL UNIQUE,
    ->     college_name VARCHAR(150) NOT NULL,
    ->     location VARCHAR(100) NOT NULL,
    ->     event_name VARCHAR(100) NOT NULL,
    ->     password VARCHAR(255) NOT NULL,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.012 sec)

MariaDB [campusconnect]> SHOW TABLES;
+-------------------------+
| Tables_in_campusconnect |
+-------------------------+
| students                |
+-------------------------+
1 row in set (0.000 sec)

MariaDB [campusconnect]> DESC students;
+--------------+--------------+------+-----+---------------------+----------------+
| Field        | Type         | Null | Key | Default             | Extra          |
+--------------+--------------+------+-----+---------------------+----------------+
| id           | int(11)      | NO   | PRI | NULL                | auto_increment |
```



```
Database changed
MariaDB [campusconnect]> CREATE TABLE students (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
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MariaDB [campusconnect]> DESC students;
+--------------+--------------+------+-----+---------------------+----------------+
| Field        | Type         | Null | Key | Default             | Extra          |
+--------------+--------------+------+-----+---------------------+----------------+
| id           | int(11)      | NO   | PRI | NULL                | auto_increment |
| full_name    | varchar(100) | NO   |     | NULL                |                |
| student_id   | varchar(50)  | NO   | UNI | NULL                |                |
| email        | varchar(100) | NO   | UNI | NULL                |                |
| college_name | varchar(150) | NO   |     | NULL                |                |
| location     | varchar(100) | NO   |     | NULL                |                |
| event_name   | varchar(100) | NO   |     | NULL                |                |
| password     | varchar(255) | NO   |     | NULL                |                |
| created_at   | timestamp    | NO   |     | current_timestamp() |                |
+--------------+--------------+------+-----+---------------------+----------------+
9 rows in set (0.001 sec)

MariaDB [campusconnect]> exit
Bye
[ec2-user@ip-172-31-18-39 ~]$
```
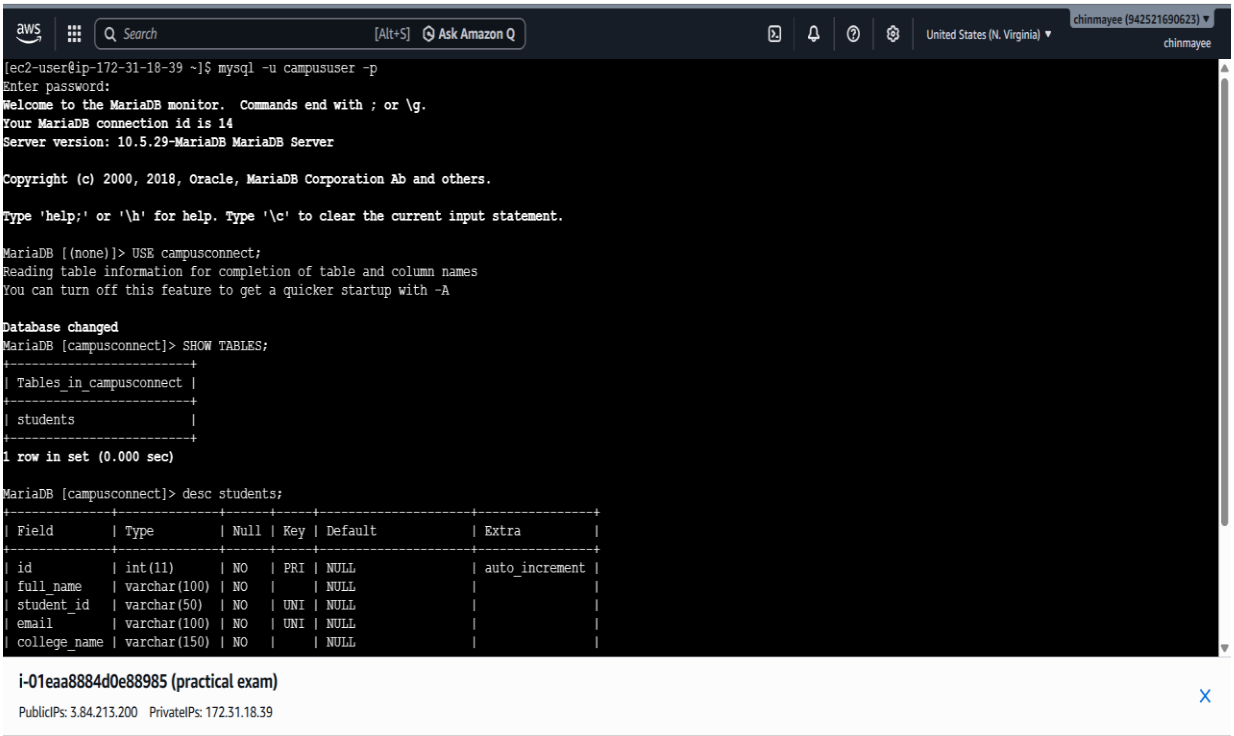
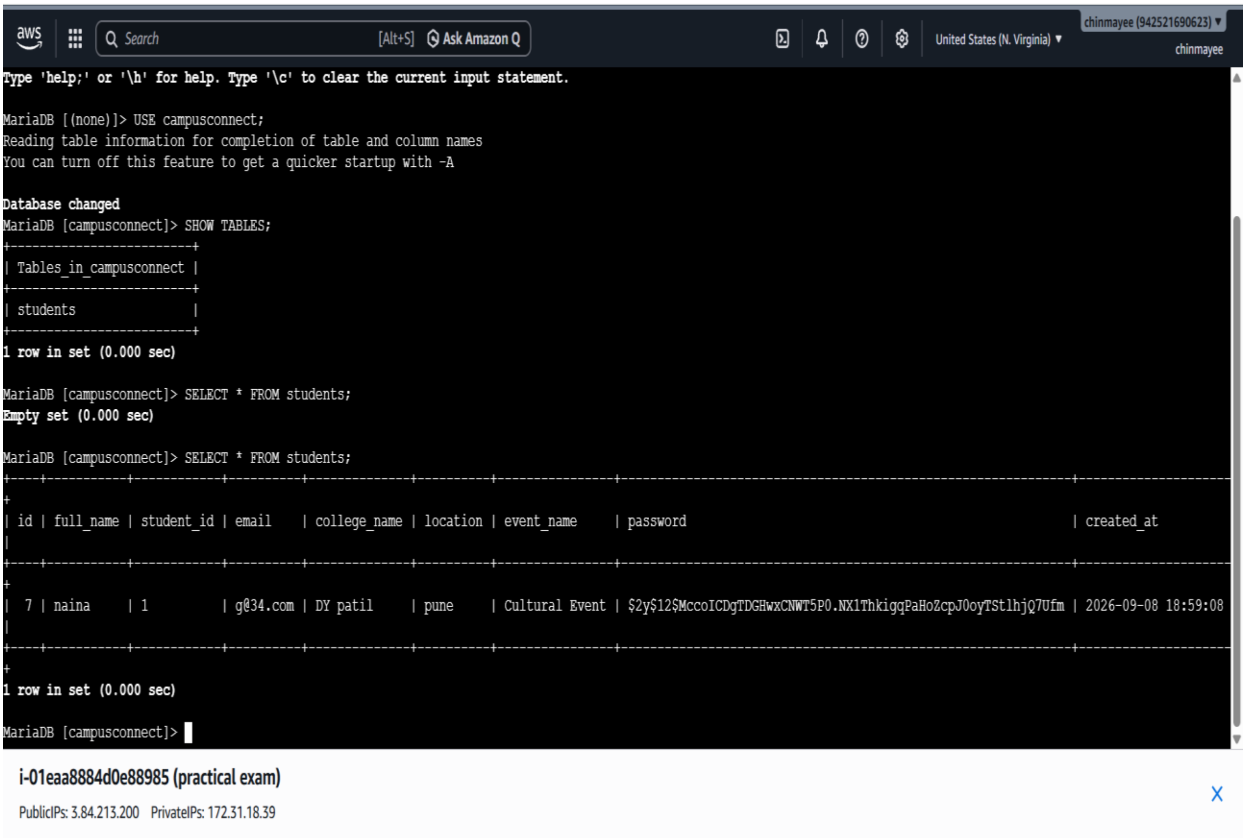
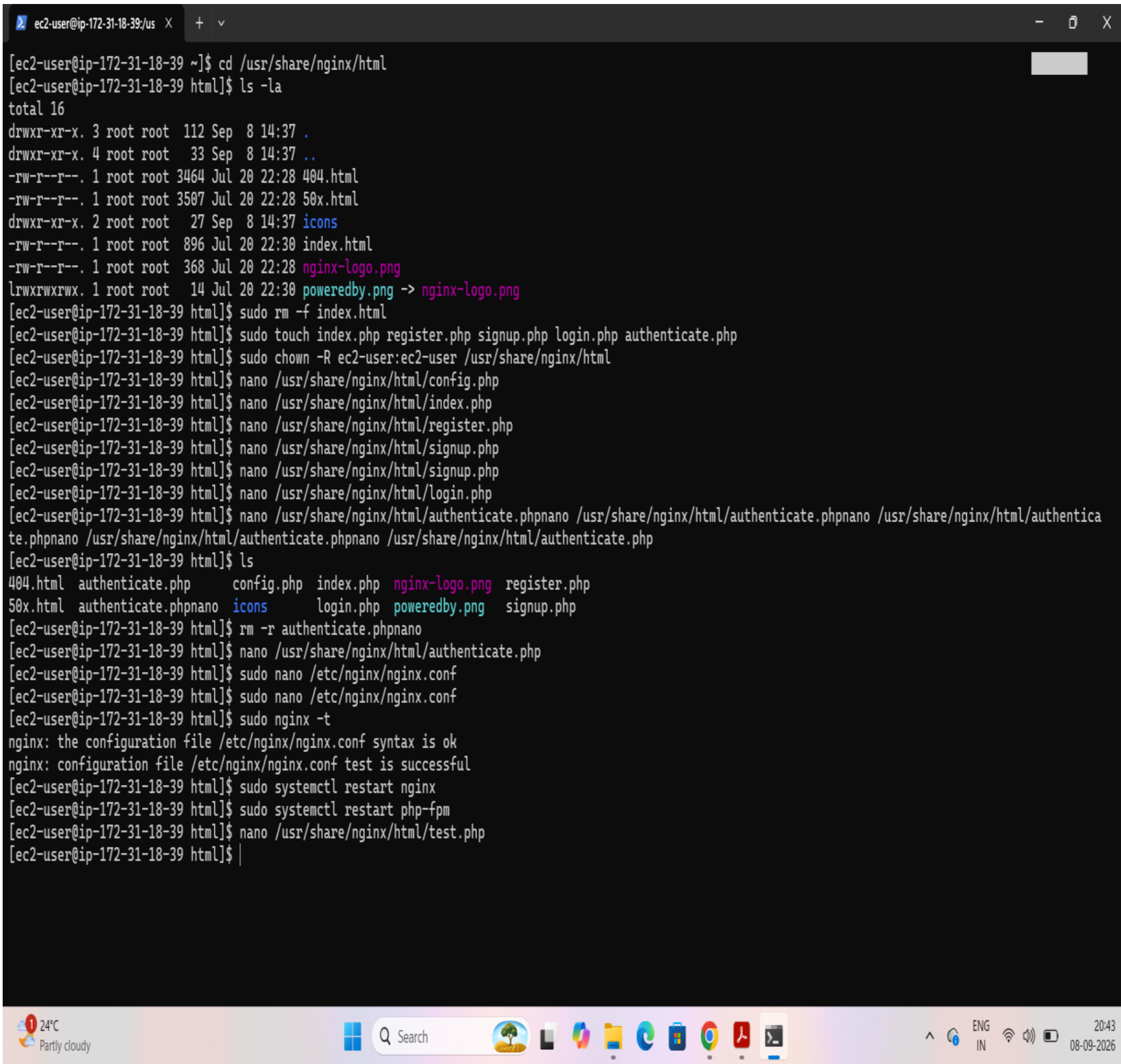

○ SQL queries ○ Inserted registration data



○ SELECT output showing registered students.

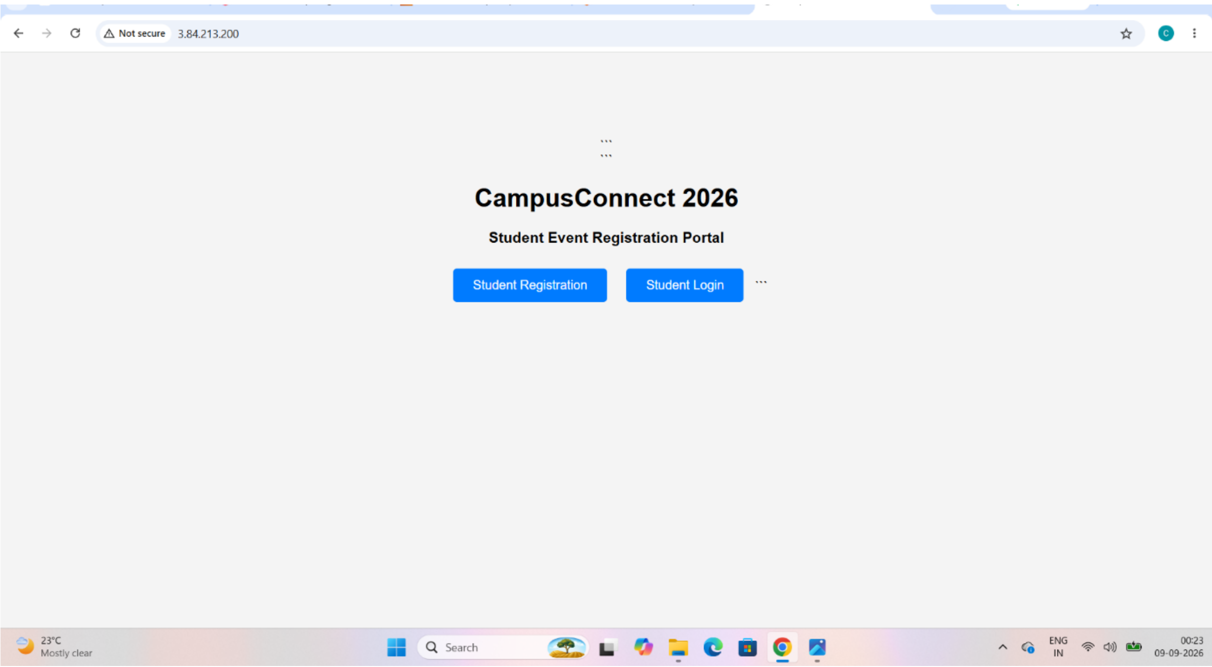


○Create Database connection files

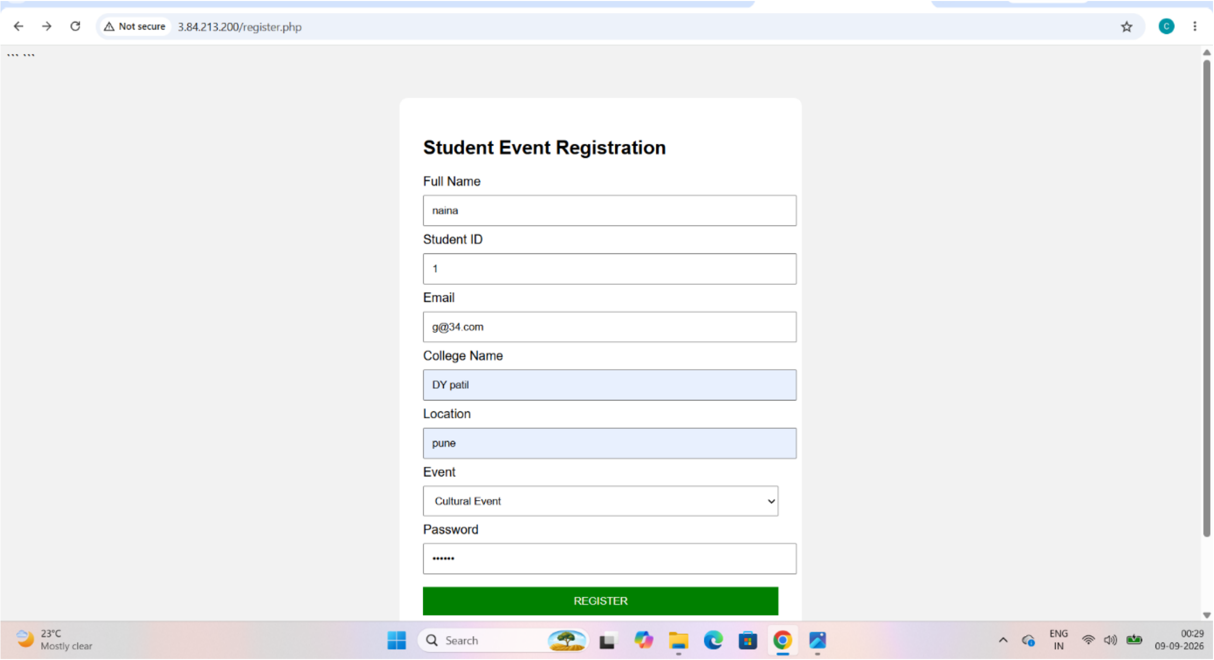


5. **Website**

- Homepage

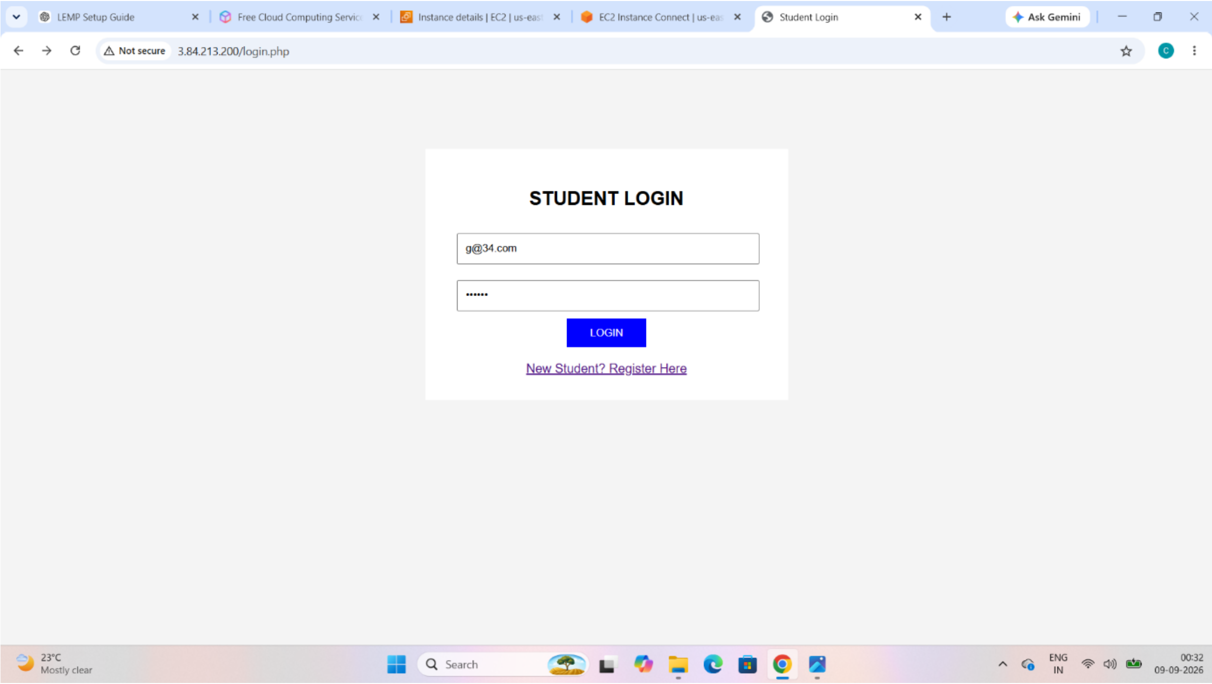


- Registration form

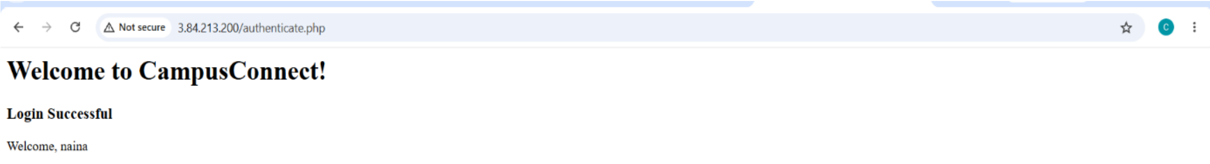


○ Successful registration

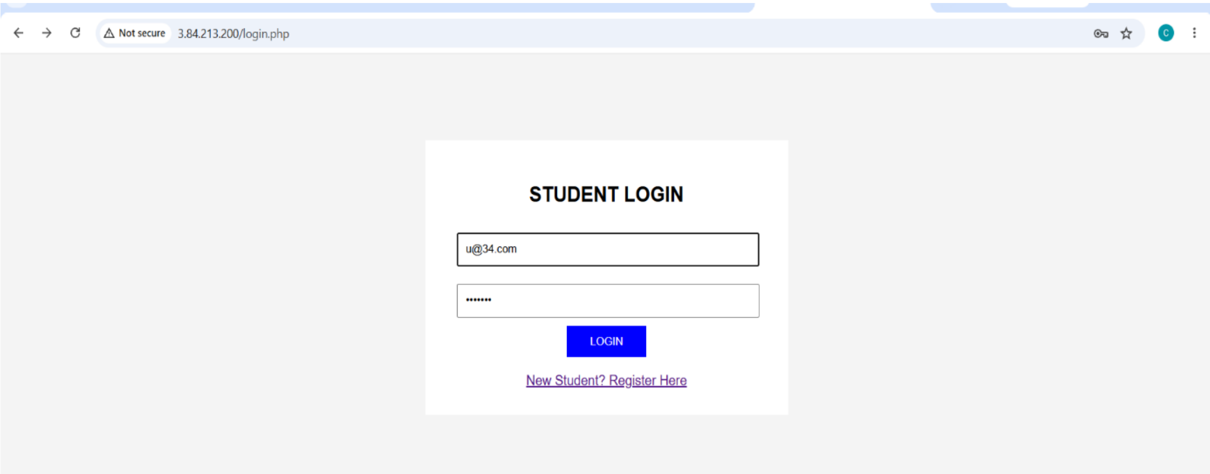
○ Login form



○ Successful login



○ Invalid login test



6. **Final Testing**
   - Service/port verification